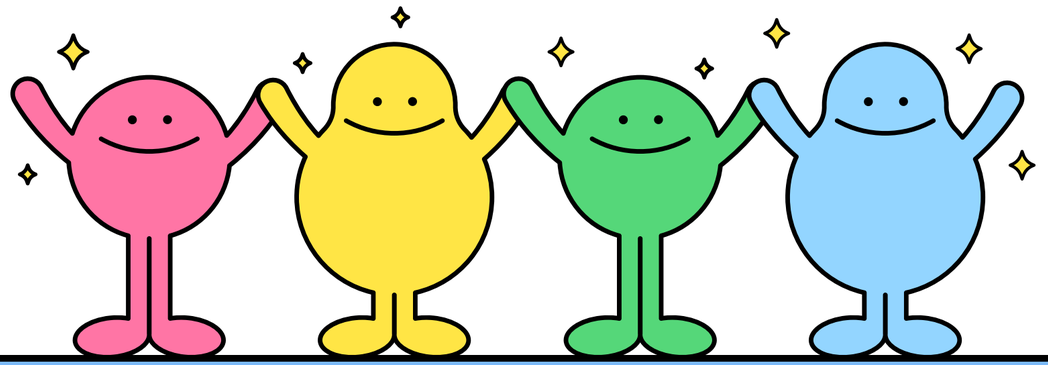
English Word Learning with Vision Object Recognition

Chalkak



21102042 Seungwoo Baik
21102052 Jeongyun Lee
21102043 Suyong Sim

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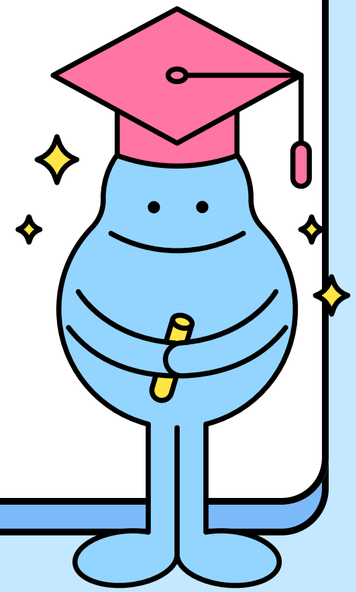
05 Technical Methodology



Background & Motivation

- Learning English words through pictures
- Users directly generate learning materials
- System for continuous learning and review
- Learn anytime, anywhere using smartphones

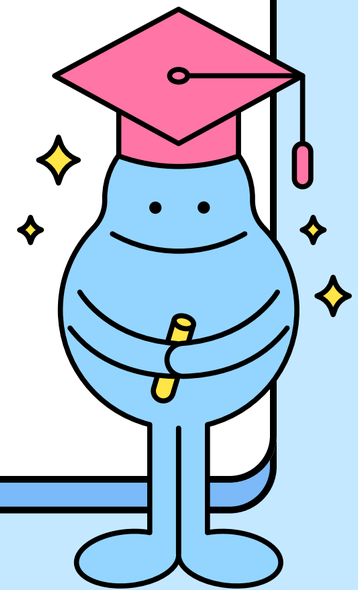






Role & Responsibility





Baik Seungwoo:

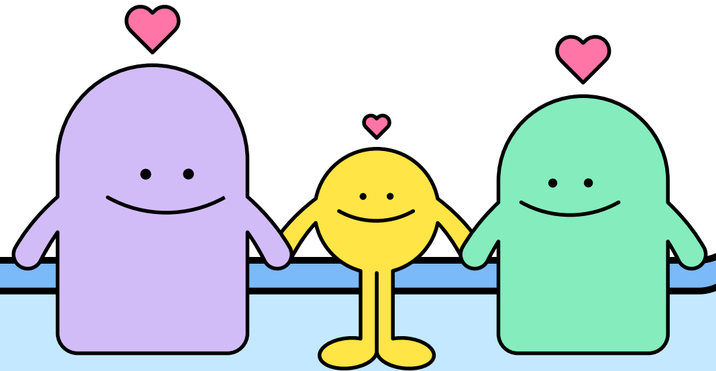
- Object Detection Model(Integrating & Using with camera/file system)
- TTS Functionality
- Sensor(accelerometer) Functionality

Sim Suyong:

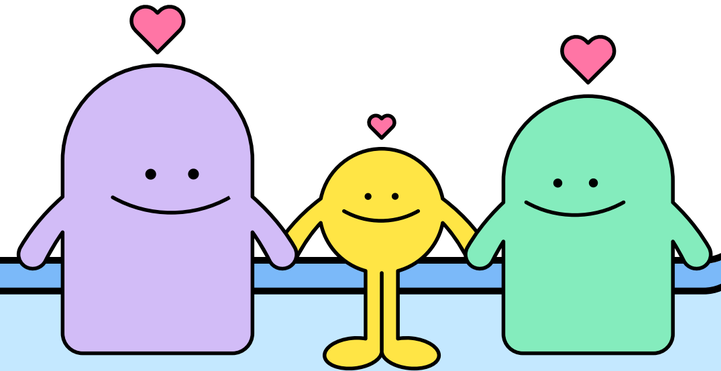
- Local RoomDB
- Open API for using ChatGPT API
- Google Firebase (Cloud server, Authentication, Notification)

Lee Jeongyun:

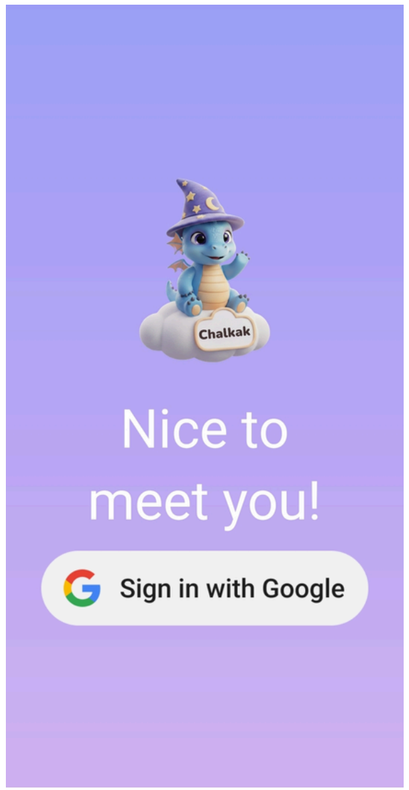
- UI/UX Implementation
- STT Functionality

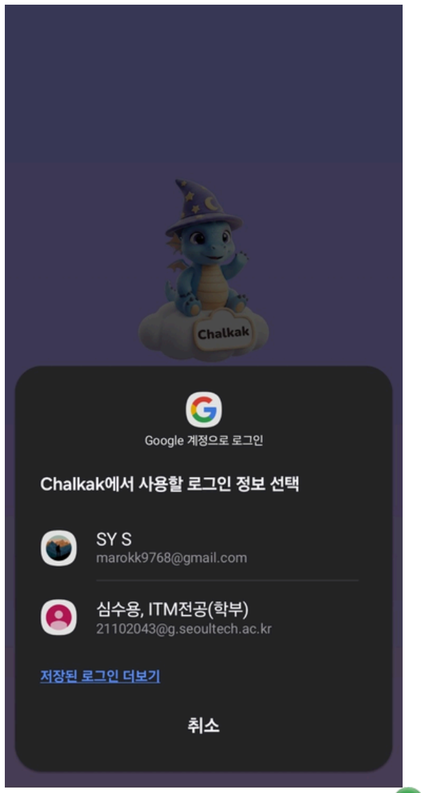


App Features



Login / Register





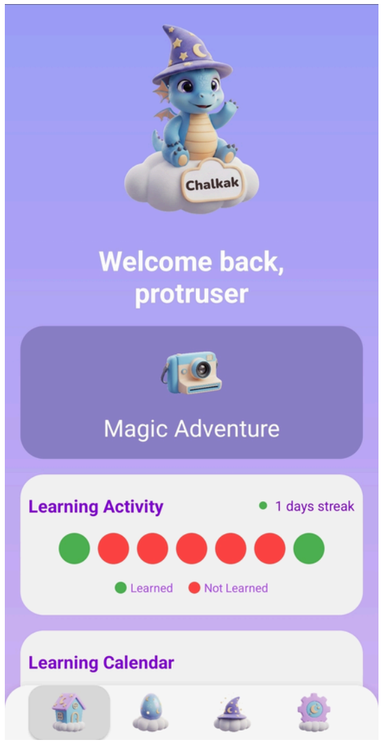
Architecture

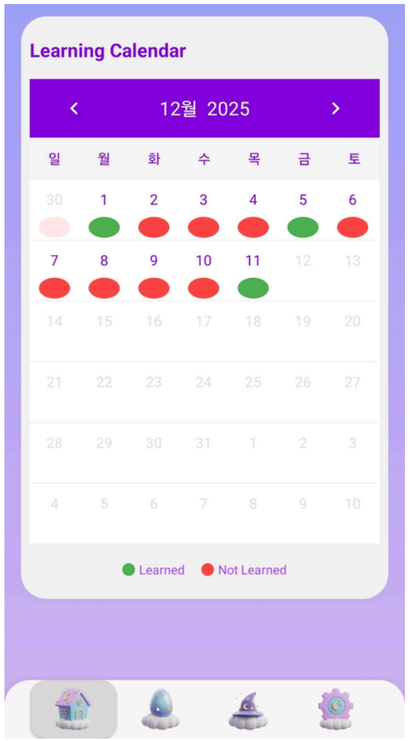
- Auth: Firebase Auth
- User DB: Firestore

Steps

1. Start login via Credential Manager / Google Sign-In
2. Authenticate with Firebase
3. Retrieve FCM Token
4. Save user info & token to Firestore
5. Go to main screen & update local/session state

Home Screen

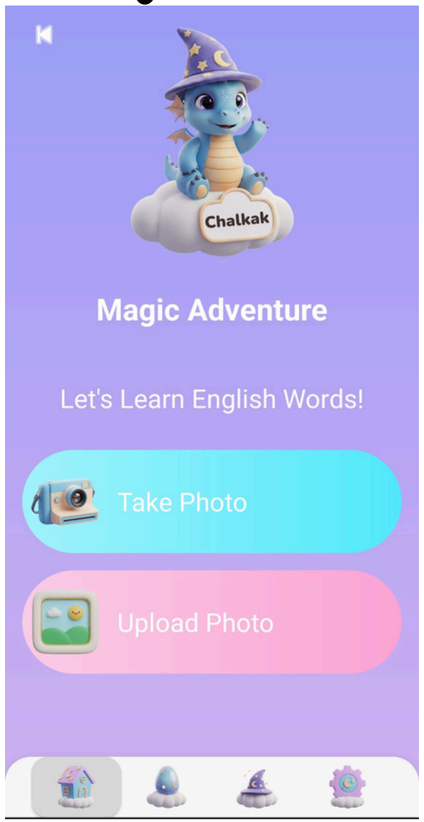


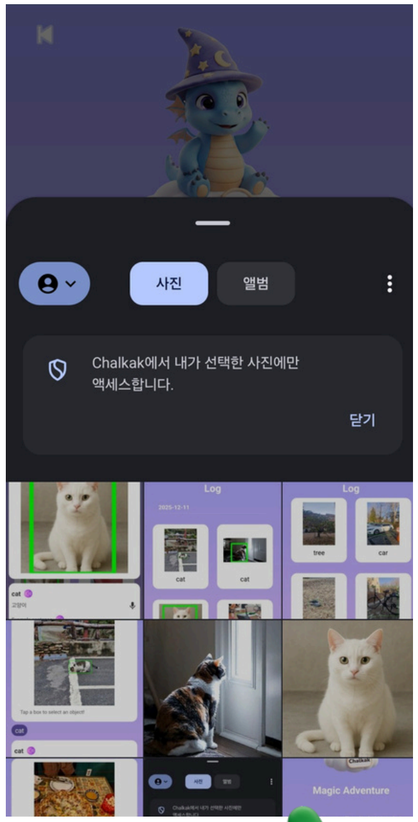


Functionalities

- Greeting with user nickname
- Access to Magic Adventure (image-based vocab)
- Last 7-day learning streak
- Calendar showing learned / not learned days
- Quick Snap access point

Learning English Words w/ Pictures (1/2)

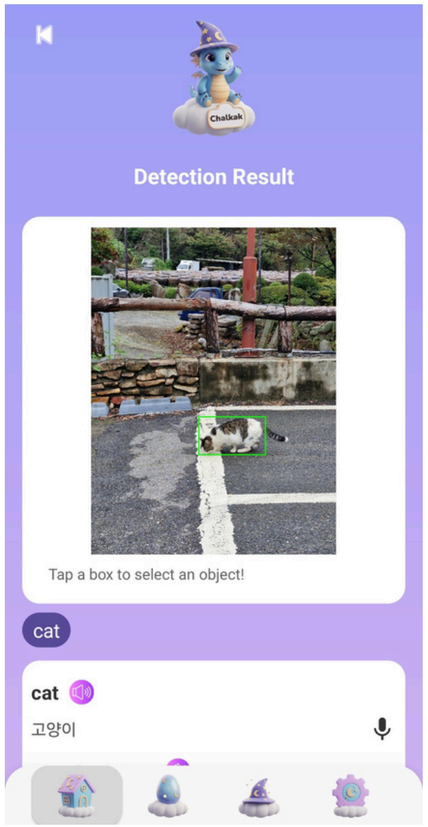


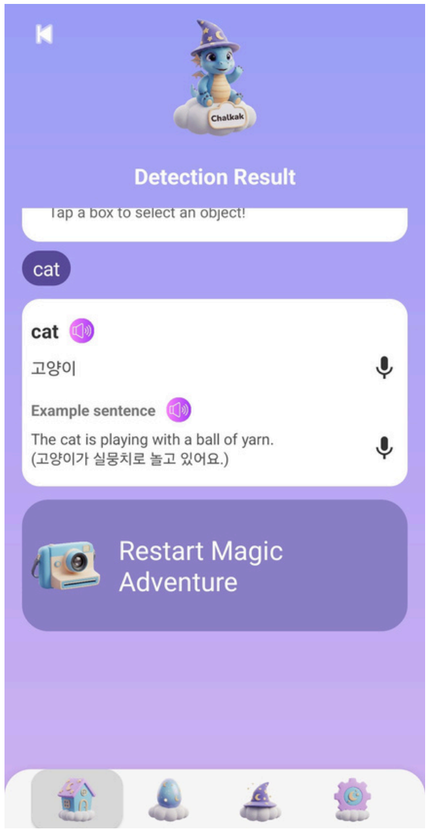


Functionalities

- Select a image by choosing Camera or Gallery Access.
- Utilize a EfficientDet-Lite0 object recognition model.

Learning English Words w/ Pictures (2/2)

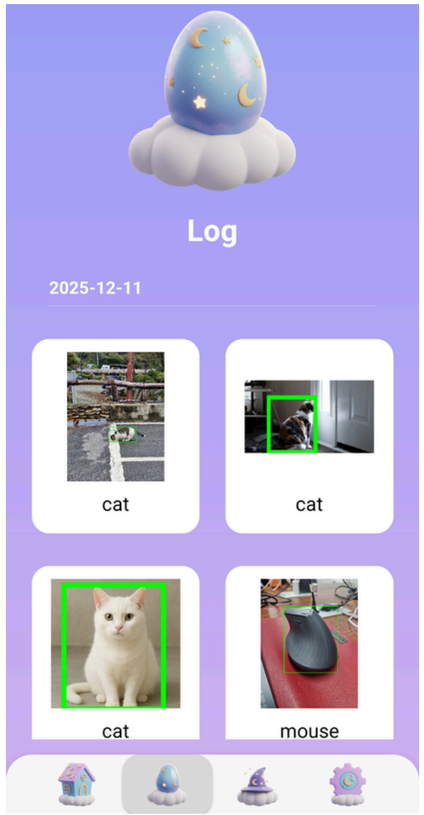


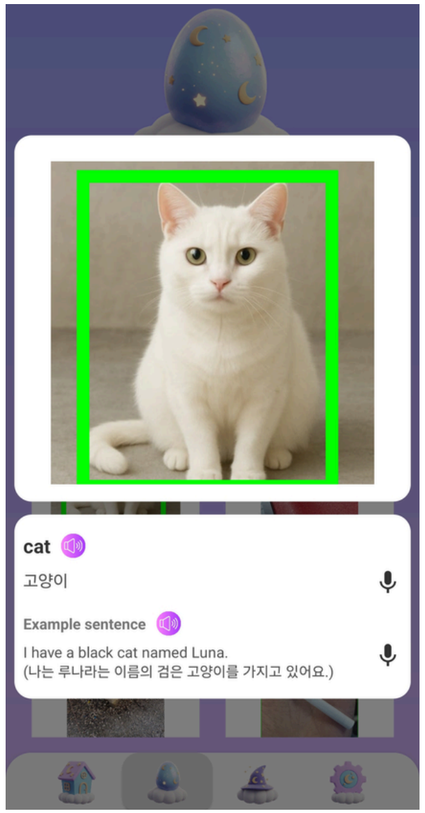


Functionalities

- Display the recognized object as a Box on the original image.
- Provide word and an example sentence with Korean meaning.
- Provide the pronunciation of the word using TTS.
- Provide user's accuracy of their pronunciation using STT.

Log

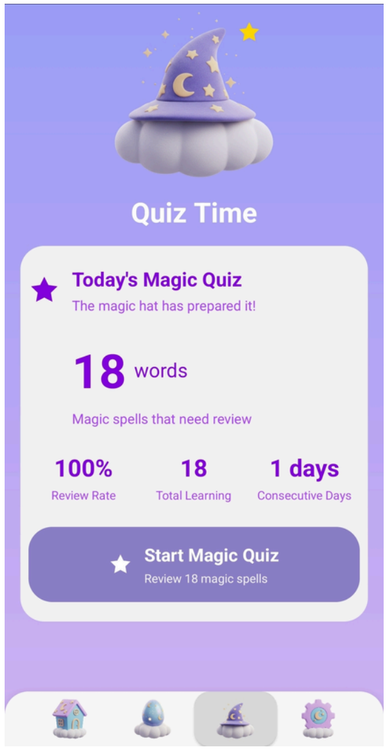


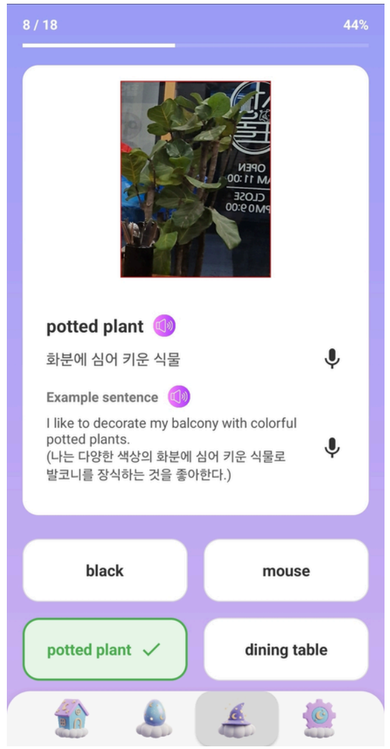


Funcionalities

- Vocabulary-photo mapping & automatic word card generation
- Multi-word detection → multi-card creation
- Grid-view cards with date-grouped organization
- Card tap → meaning + example sentences
- TTS/STT for pronunciation and speaking practice

Quiz

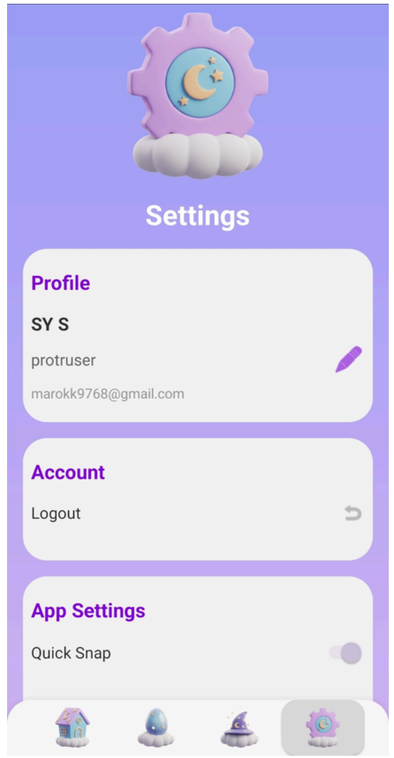


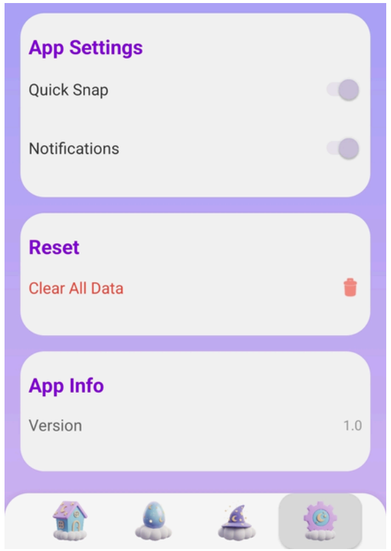


Functionalities

- Daily target vocabulary count
- Quiz attempt count & overall review rate
- Consecutive quiz-day streak
- Four-choice quiz format with instant feedback
- Correct answer → word, meaning, example sentence, TTS/STT
- Progress indicator at the top during quizzes

Settings



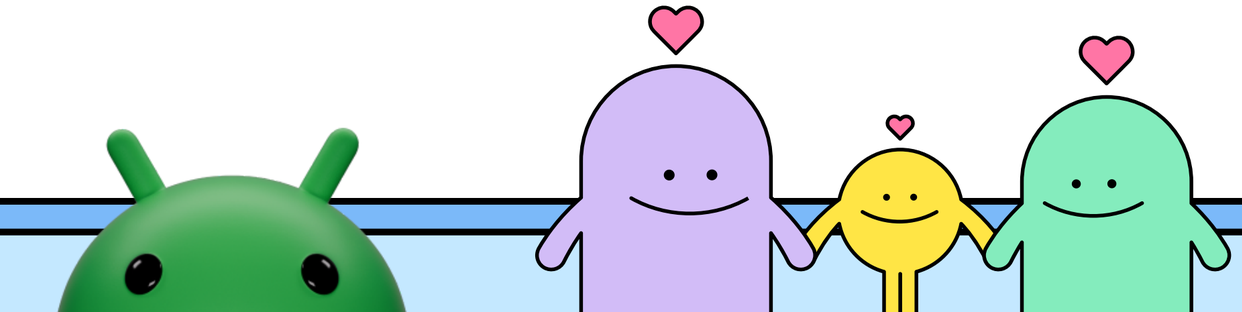


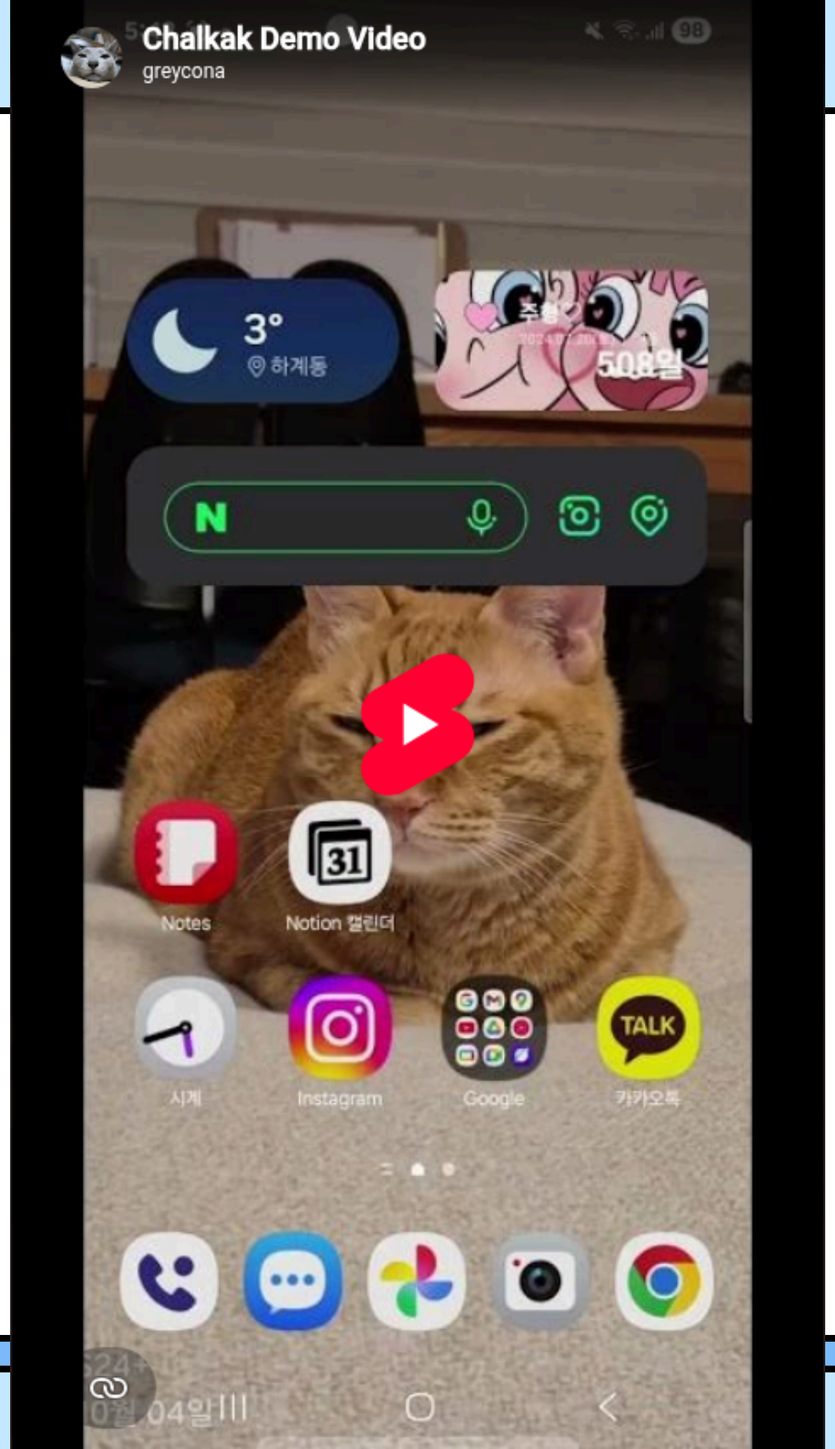
Functionalities

- Nickname customization
- Logout option
- Quick Snap toggle
- Notification toggle
- App data reset function

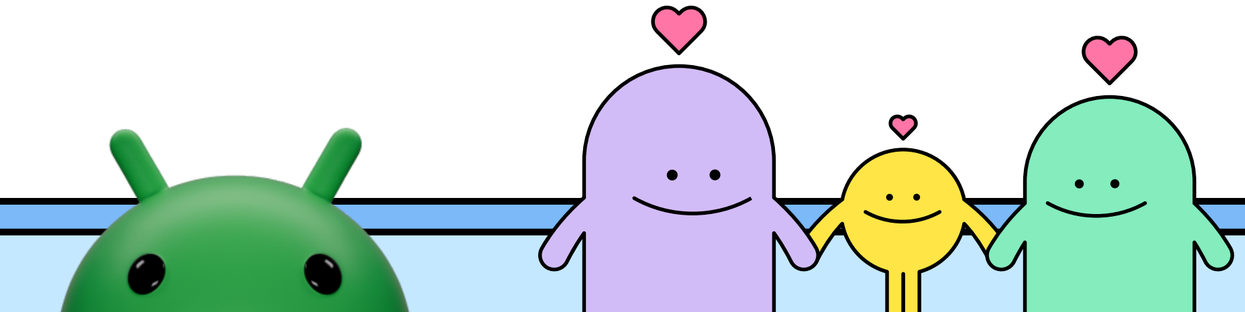


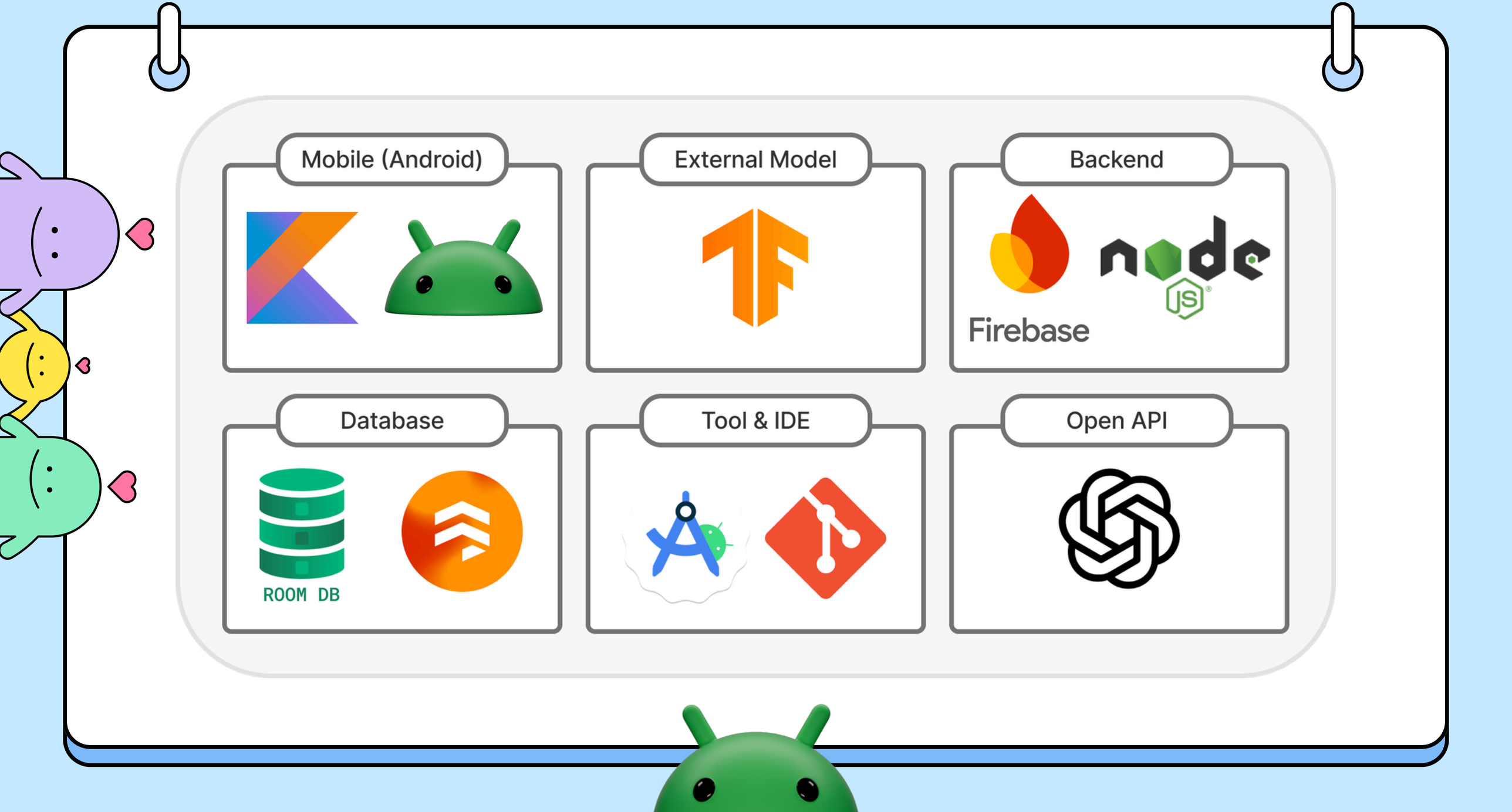
Demonstration Video



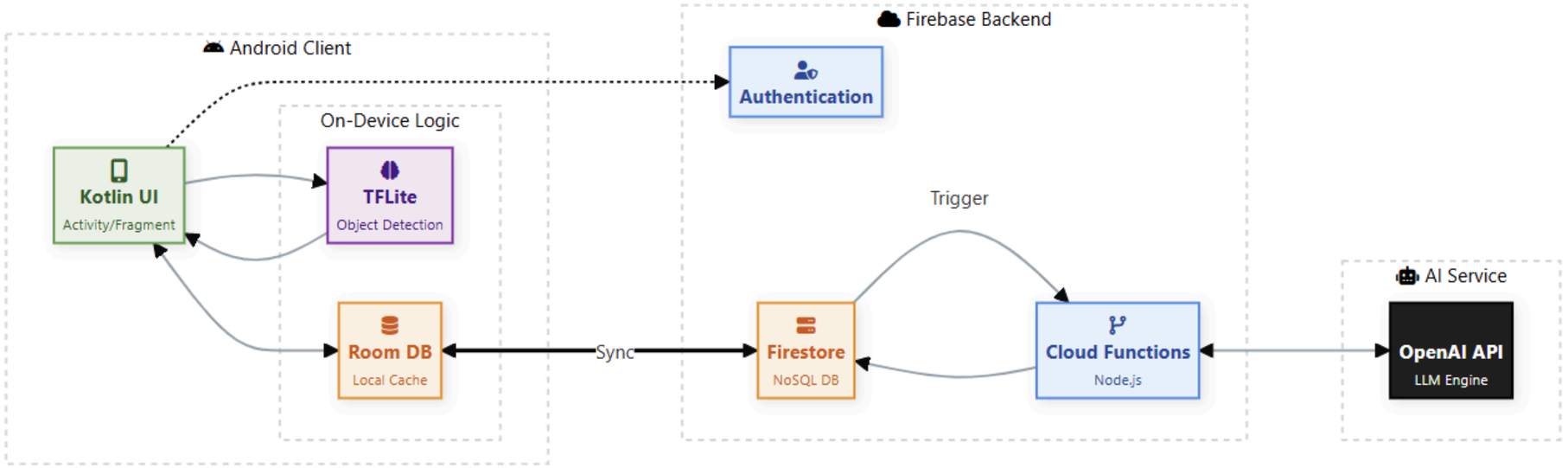

Chalkak Demo Video
greycona
3°
하계동
508일
Notes
Notion 캘린더
시계
Instagram
Google
카카오톡
04일

Technical Methodology

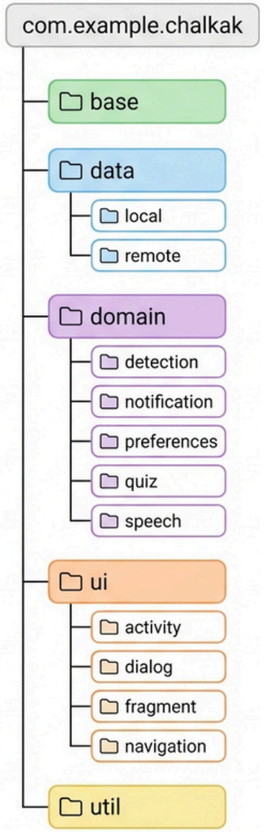



Mobile (Android)
External Model
Backend
Firebase
Database
ROOM DB
Tool & IDE
Open API

Architecture Flow



Mobile (Android) - File Structure



File Structure

- base/: Collection of common base classes or shared abstractions (Providing common logic for Fragments).
- data/: External and internal data source layer.
  - local/: On-device storage like Room.
  - remote/firestore/: Network/backend integration (e.g., Firestore).
- domain/: Core domain logic and feature-specific helpers. Non-UI logic such as Detection, Notification, Settings, Quiz, and Voice.
- ui/: Screen layer (User Interface).
  - activity/, fragment/, dialog/, navigation/
- util/: General-purpose utilities (e.g., Image loading/selection, Sensor, Toast, Data loading, etc.).





Mobile (Android) - UI Architecture

App Architecture

- Uses a classic single-activity, multi-fragment architecture.
- MainActivity acts as the main container: bottom navigation, global UI, and window insets handling.
- Core screens like Home, Log, Quiz, Settings, and Magic Adventure are implemented as Fragments.

UI Pattern

- Activity: Shell of the app, global navigation, and external entry points (e.g., main screen, image preview).
- Fragment: Main screen units under bottom navigation, plus swappable sub-screens.
- DialogFragment: Temporary overlay for details or secondary actions (e.g., full-screen word detail with TTS and speech input).
- Built with XML layouts + classic View system + findViewById.
- Screen transitions use manual FragmentTransactions.





Mobile (Android) - TTS / STT

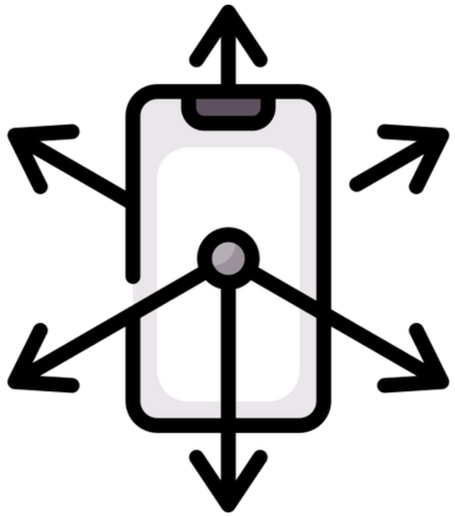
TTS

- Use Android's built-in TextToSpeech to set up an English voice (slower rate).
- Wire the word/example TTS buttons on the card view to init.
- On screen destroy, stop and shut down the TTS engine to release resources.

STT

- Use Android's built-in SpeechRecognizer and check mic permission
- If granted, start listening with an english - -US intent.
- Grab the two mic icons (word/example) and the texts (selected word, example) on the card on button tap, toggle listening and swap the icon between mic and stop.
- When recognition returns, compare it with the original English word and show whether it matches.

Mobile (Android) - Sensor



- Sensor Registration
  - Register listener to accelerometer via SensorManager.
- Acceleration Measurement
  - Calculate G-Force from X, Y, Z axis values.
- Shake Detection
  - Trigger when G-Force > 2.0 G
- Camera Launch
  - Execute callback to open camera screen.



Mobile (Android) - Camera Access

Common Mechanism: URI--based Flow

- Uses the Activity Result API: call external component → get result via callback.
- Both camera and gallery ultimately return an image Uri.
- Downstream screens load and display the image using this Uri.

Camera Flow

- The app pre-creates an empty image file in its internal cache directory.
- That file is wrapped as a content:// Uri via FileProvider and passed to the camera app.
- The camera writes the captured photo directly to that Uri location.
- On success, the callback receives the already-saved image Uri.
- Because it lives in the app's internal cache, the app can access the photo without relying on the system gallery.



Mobile (Android) - Gallery Access

Common Mechanism: URI-based Flow

- Uses the Activity Result API: call external component → get result via callback.
- Both camera and gallery ultimately return an image Uri.
- Downstream screens load and display the image using this Uri.

Gallery Flow

- A system picker is launched in “content request” mode (e.g., GetContent("image/*")).
- The OS returns a Uri that represents the image selected by the user.
- The app gets read access to that Uri without needing the actual filesystem path.
- Conceptually, the app only obtains a handle to an already existing file and uses it by reference.



Mobile (Android) – Calendar View



Uses Material CalendarView to visualize learning activity with dots on each date.
Fetches learning logs from a Room database and aggregates them by date.

Date Status Visualization

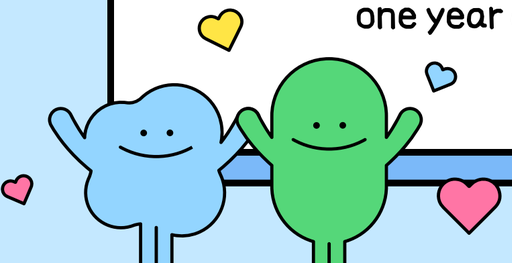
- Green dot: days with learning activity based on photo logs.
- Red dot: days with no learning between the first learning date and today.
- Calendar range is limited to the last 1 year (from one year ago to today).

User Interaction

- When a user taps a date, the app checks if it's a "learned" or "not learned" day.
- Shows a toast message: "Completed" or "No learning" accordingly.

Activity Metrics & Streaks

- Reuses the same learning logs to calculate streaks and recent activity.
- Displays current streak, recent 7-day activity, and small activity dots in the header area.







External Model (1/2)

Tensorflow - EfficientDet Lite0 model

Pre-trained TensorFlow object detection model trained on the COCO dataset.

- Model Loading & Utilization:
  - The model is loaded and utilized by using the methods of the classes provided by TensorFlow.
- Helper Class Implementation:
  - A dedicated Helper class was implemented for model loading and initialization.
  - Maximum number of objects to recognize: 2
  - Minimum confidence threshold for object recognition: $0.3f

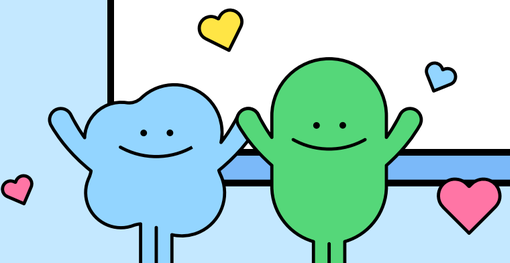


External Model (2/2)





- Model Output/Results:
  - The model returns results as a list containing the following information for each detected object:
    - Recognized Object (Class Label)
    - Confidence Score
    - Position of the object within the image
    - The image with a bounding box drawn around the detected object's location.
- Data Processing and User Delivery:
  - The returned results are stored in the database (DB).
  - The results are then provided to the user as part of the results screen.







Backend(1/2)

Firebase

Authentication Flow (Google OAuth 2.0)

Flow: Credential Manager → Google ID Token → Firebase Auth → Session

Mechanism:

- Credential Manager: Initialize an OAuth request through Android Credential Manager API with Authentication Intent
- Google ID Token: Receive Custom Credential result from the system, extract OAuth 2.0 token, and prepare server-side verification
- Firebase Auth: Convert ID token into Google Auth Provider credential and let Firebase Authentication validate the token signature with Google server
- Session: After verification, send UID and email to firestore and make session

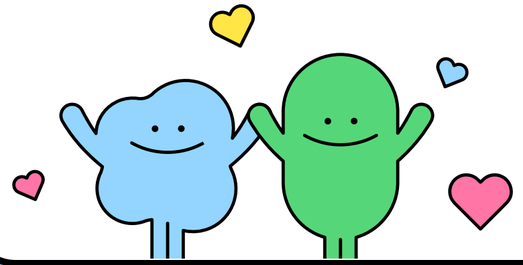
Backend(2/2)

Push Notification

Flow: Token Generation → Firestore (Mapping) → Scheduler/Event → User Device

mechanism:

- Token Generation: App startup triggers the FCM SDK to create a unique device registration token.
- Firestore (Mapping): The token is saved to the user's Firestore document to bind the account to the device.
- Scheduler / Event:
  - Local: System alarms fire reliably even in Doze(절전 모드)
  - Remote: Server looks up the token and targets the specific device
- User Device:
  - Local: A system handles the alarm and shows the notification
  - Remote :A background service processes FCM payloads and wakes the app







Database

Local Database
Android Room

PhotoLog — TABLE (PARENT)

🔑 photoId	Long (PK)
localImagePath	String
createdAt	Long (Timestamp)

↓ 1 : N

DetectedObject — TABLE (CORE)

🔑 objectId	Long (PK)
🔗 parentPhotoId	Long (FK)
🔄 englishWord	String (Index)
koreanMeaning	String
boundingBox	String "[l,t,r,b]"
lastStudied	Long

↓ 1 : N

ExampleSentence — TABLE (CHILD)

🔑 id	Long (PK)
🔗 wordId	Long (FK)
sentence	String
translation	String

Sync Key: "English Word"

User ID

Remote Database
Cloud Firestore (NoSQL)

users — COLLECTION

Document ID	Auth UID
email	String
nickname	String
fcmToken	String
lastStudiedAt	Timestamp/String
stats	Map { totalWordCount: Int }

words — GLOBAL CACHE

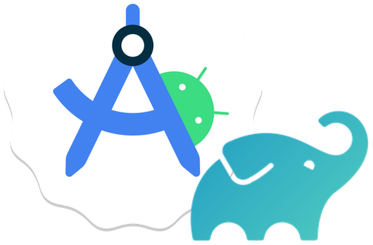
Document ID	English Word (e.g. "apple")
originalWord	String
meaning	String
examples	Array of Maps

- sentence: String
- translation: String



Tools and IDE





Android Studio

- Development based on Android 13.0, specifically targeting the S23 FE.

Gradle

- Controll the dependencies to integrate TensorFlow Lite, Room, and Firebase

Git

- Git setup for version control and collaboration
- Integration with a GitHub Organization
- Collaboration is managed through branches and GitHub Issues.
  - https://github.com/chalkak-itm
  - https://github.com/chalkak-itm/chalkak-app.git







Open API



GPT Integration

flow: Client App → Cloud Function → GPT-3.5-turbo → Client/DB

mechanism:

- Secure Endpoint: Implemented as a Firebase Function
- JSON Enforcement: Parse the response directly as strict JSON structure
- Automated Sync: The parsed data is persisted to RoomDB and Firestore simultaneously via Dual-Write





Open API (JSON Format)



System : You are a helpful English-Korean dictionary assistant. Provide Korean translations and example sentences in JSON format.

```
{
  "meaning":"고양이",
  "examples":[
    {
      "sentence":"I have a black cat named Luna.",
      "translation":"나는 루나라는 이름의 검은 고양이를 가지고 있어요."
    },
    {
      "sentence":"The cat is playing with a ball of yarn.",
      "translation":"고양이가 실뭉치로 놀고 있어요."
    },
    {
      "sentence":"She loves to cuddle with her cat on the couch.",
      "translation":"그녀는 소파에서 그녀의 고양이와 껴안는 것을 좋아해요."
    }
  ]
}
```





Thank you
for listening



TEAM Chalkak

